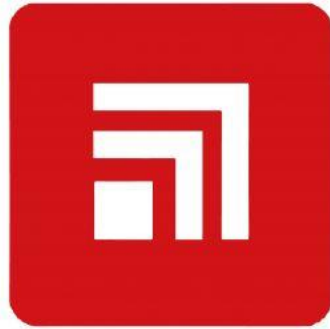


**Ethical Challenges in AI-Powered Loan Approval System
Case Study Analysis**



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1. Key Ethical Issues in the Case

- The case of FutureTrust Bank highlights several important ethical concerns arising from the use of AI in financial decision-making.

Firstly, algorithmic bias and discrimination is clearly visible. Female applicants and individuals from Tier-2 and Tier-3 cities are being rejected more often, even when their financial profiles are similar to others. This indicates that the system is not treating all applicants fairly.

Secondly, there is a lack of transparency, often referred to as the “black box problem.” Customers are not given proper reasons for rejection and only receive a generic response like “system decision.”

Another issue is unfair decision-making, where individuals with similar qualifications are treated differently.

There is also a lack of accountability, making it difficult to identify responsibility for incorrect decisions.

Additionally, the system suffers from data ethics issues, as it relies on historical data that already contains biases.

Lastly, these issues collectively damage customer trust and brand reputation.

2. Reasons Behind Unfair Outcomes

- The unfair outcomes produced by the AI system can be attributed to multiple factors.
- The most significant reason is biased training data. Since the model was trained on historical data, it learned patterns that may reflect past discrimination.
- Another reason is the confusion between correlation and causation.
- The system was also designed with a focus on accuracy rather than fairness.
- Moreover, the absence of explainability tools prevented developers from identifying biased patterns.

Finally, there was a lack of continuous monitoring and auditing.

3. Is Model Accuracy Alone Sufficient?

Model accuracy alone is not sufficient to evaluate success, especially in sensitive domains like banking.

A model can achieve high accuracy overall but still perform poorly for specific groups.

Factors such as fairness, transparency, reliability, and ethical compliance must also be considered.

4. Role of Explainability in Improving Customer Trust

Explainability plays a crucial role in building customer trust.

It helps customers understand why decisions are made, reduces perceptions of bias, and improves transparency.

It also enables better internal monitoring and supports regulatory compliance.

5. Accountability for Wrong AI Decisions

- Accountability should be shared across multiple stakeholders.
- Bank management is responsible for implementation and governance.
- Data scientists and developers are responsible for model design and training.
- Business decision-makers define objectives, and compliance teams ensure regulatory adherence.

Human accountability must always exist.

6. Suitable Fairness Metrics

- Demographic Parity ensures equal approval rates.
- Equal Opportunity ensures fairness among qualified applicants.
- Disparate Impact Ratio measures discrimination.
- Error Rate Balance compares mistakes across groups.
- Calibration ensures reliability of predictions.

7. Recommendations to Improve the System

- Improve data quality by removing biases.
- Introduce fairness constraints in the model.
- Implement explainable AI tools.
- Adopt a human-in-the-loop approach.
- Conduct regular ethical audits.
- Improve customer communication.
- Develop a strong AI governance framework.

8. Should Loan Approvals Be Fully Automated?

- Loan approvals should not be fully automated.
- AI improves efficiency but lacks human judgment.
- A hybrid approach combining AI and human review is more effective and ethical.

Conclusion

The case shows that AI improves efficiency but introduces ethical challenges. A responsible system must ensure fairness, transparency, and accountability. A balanced approach combining technology and human oversight is essential.